CYB102 Milestone 1

(🔗 Instructions Page)

Team Members (Required)

Reminder: Make sure to provide **edit access** for this Milestone document to **everyone on your team!**

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☕Favorite Drink:	Hot chocolate

What are pronouns / Why are they included here?

Select one (or more) open-source Datasets to analyze (Required)

Data Set Chosen: The data set we have chosen to analyze for The Data Dig is...

Name:	SolarWinds IOC Dataset + Network Proxy Log Dataset
Primary Link:	https://raw.githubusercontent.com/codepath/cyb102-file-storage/main/SolarWindsIOCs.csv
Other Resource:	https://raw.githubusercontent.com/codepath/cyb102-file-storage/main/NetworkProxyLog02.csv
Other Resource:	

Data Set Description: Where does the data come from? Who generated it? What kind of devices / technologies does it target? What format is the data in?

This project uses two datasets: a SolarWinds IOC dataset containing known malicious indicators and a network proxy log dataset containing internal network events. The IOC dataset includes suspicious IP addresses associated with the SolarWinds campaign, while the proxy log dataset includes timestamps, IP addresses, and computer names from observed network traffic. By comparing the two datasets, we can identify systems that communicated with known malicious infrastructure.

Hypothesis: What are 3 things you expect to find when you analyze the data?

Tip: You won't lose points if these hypotheses turn out to be wrong! Make educated guesses!

Finding #1:	We expect to find internal systems that communicated with IP addresses known to be associated with the SolarWinds campaign.
Finding #2:	We expect that some of the matched IP addresses will also be flagged as malicious or suspicious by external threat intelligence sources such as VirusTotal.
Finding #3:	We expect to find that more than one internal system was affected, which would suggest a broader impact across the network rather than a single isolated event.

Select an incident-response playbook to follow (Required)

Playbook Chosen: The playbook we have decided to follow for The Data Dig is...

Name:	NIST Incident Response Framework
Primary Link:	https://nvlpubs.nist.gov/nistpubs/SpecialPublications/NIST.SP.800-61r2.pdf
Other Resource:	
Other Resource:	

Playbook Description: Who wrote this playbook? Who is the target audience? Does it make any specific assumptions about the data set? If so, do those match your data, or will you have to adapt the playbook?

The NIST Incident Response Framework provides a structured process for handling cybersecurity incidents through preparation, detection and analysis, containment, eradication, and recovery. It fits this project because the team is detecting suspicious events, analyzing potential compromise, validating indicators with threat intelligence, and recommending remediation steps.

Tools we Plan to Use: Based on your dataset and playbook, what blue-team tools from this course will you use to analyze the incident? (MINIMUM of 2)

Tool #1:	Splunk
Tool #2:	VirusTotal
Tool #3:	
Tool #4:	
Tool #5:	

Answer each of the *key aspect* questions (Required)

Instructions: *For each of the key aspects below, include a few sentences explaining how your project is demonstrating that aspect. Please include at least one specific example.*



For a full definition of each of the key aspects, please view the Data Dig Project page on the Course Portal.

Monitoring Sources

How it relates to our project:	Monitoring sources in this project include the network proxy logs and the SolarWinds IOC dataset. The proxy logs provide visibility into internal network activity, while the IOC dataset provides known malicious indicators. Together, they allow us to detect suspicious communication between internal systems and external malicious infrastructure.
Example(s):	The proxy log data showed that internal systems were connecting to external IP addresses. By comparing these IPs with the SolarWinds IOC list, we identified matches such as 5.252.177.25 and 13.59.205.66, indicating suspicious activity.

Identified Assets

How it relates to our project:	Identified assets in this project are the internal systems that communicated with suspicious IP addresses. These systems are important because they may have been exposed to malicious infrastructure and could be compromised.

Example(s):	The system LN-SolarStrike-14 communicated with the IP 5.252.177.25, and WS-SolarWave-212 communicated with 13.59.205.66. These systems are considered affected assets and should be prioritized for investigation.

Impact Analysis and Triage

How it relates to our project:	Impact analysis and triage in our project focus on determining how severe the potential threat is and which systems were affected. By matching IP addresses from the network proxy logs against known SolarWinds IOCs, we identified internal systems that communicated with suspicious external infrastructure. This helps us prioritize response actions by showing which assets may have been exposed and which events require immediate investigation.
Example(s):	The IP address 5.252.177.25 appeared in both the network logs and the SolarWinds IOC dataset and was associated with the system LN-SolarStrike-14 at 2024-03-05 07:11:28. Another match was 13.59.205.66, which was associated with WS-SolarWave-212 at 2024-03-04 06:57:28. Because both IPs were also flagged as malicious in VirusTotal, these findings increase the severity of the incident and suggest that the affected systems should be prioritized for containment and further investigation.

Threat Intelligence

How it relates to our project:	Threat intelligence is used to validate whether identified IP addresses are associated with known malicious activity. By using external sources such as VirusTotal, we can determine if the matched IPs pose a real threat.
Example(s):	VirusTotal analysis showed that both 5.252.177.25 and 13.59.205.66 were flagged as malicious by multiple security vendors. This confirms that these IPs are associated with suspicious activity and strengthens the investigation findings.

Recommended Remediation

How it relates to our project:	Recommended remediation focuses on reducing risk and preventing further communication with malicious infrastructure. This includes blocking malicious IPs and investigating affected systems.
Example(s):	The IP addresses 5.252.177.25 and 13.59.205.66 should be blocked at the firewall level. Additionally, affected systems such as LN-SolarStrike-14 should be scanned for malware and monitored for suspicious behavior.

Case Management System (and screenshots)

How it relates to our project:	Case management involves organizing and tracking all investigation findings, including matched IP addresses, affected systems, and timestamps. This ensures the incident response process is structured and documented.
Example(s):	The matched IPs, associated systems, and timestamps can be documented in a case report. This allows the team to track progress, assign tasks, and ensure proper remediation steps are taken.

Presentation Prep (Required)

Presentation Plan: What is your plan for the presentation? Please include a roadmap, flowchart, diagram, or outline.

Things to consider:

- ☑ ~~What will you talk about, and in what order?~~
- ☑ ~~Who will be talking at what times?~~
- ☑ ~~What visual-aids will you use?~~

Presentation Plan

1. Introduction (Speaker 1 – ~1 min)
Brief overview of the topic and why it matters
State the objective of the presentation
Preview what the audience will learn

2. Assignment Overview (Speaker 1 – ~1–2 min)
Describe the problem or scenario
Define scope and goals of the investigation

3. Dataset & Tools (Speaker 2 – ~2 min)
Introduce dataset(s) used (e.g., logs, sources)
Explain tools/platforms (e.g., Splunk, etc.)
Visual Aid: Screenshot or diagram of tool interface

4. Investigation Methodology (Speaker 2 – ~2–3 min)
Explain step-by-step approach taken
Justify why this method was the best choice
Visual Aid: Flowchart of investigation process

5. Findings (Speaker 3 – ~3 min)
Key results from analysis
Highlight important patterns, anomalies, or evidence
Visual Aid: Charts, tables, or query results

6. Impact Analysis (Speaker 3 – ~2 min)
Explain the significance of the findings
Potential risks or consequences
Who/what is affected

7. Threat Intelligence & Remediation (Speaker 4 – ~2–3 min)
Connect findings to known threats (IOCs, attack types)
Provide recommended mitigation strategies
Visual Aid: Bullet summary or threat mapping

8. Playbook & Case Management (Speaker 4 – ~2 min)
Describe response steps (incident handling process)
Show how the case would be managed in real-world scenarios
Visual Aid: Workflow diagram

9. Conclusion (Speaker 1 – ~1 min)
Summarize key takeaways
Reinforce importance of findings
End with final insight or recommendation

Stretch Feature: Custom Playbook (Optional)

If you have chosen to write or modify a playbook, document it here.

Tip: To link your drafts, we recommend using Google Drive files. **Be sure any linked files are set to *"Anyone with the link can View"*!** If the grading team cannot open your file, you **will not get credit** for this stretch feature.

Original Playbook: The original playbook we started with / used as inspiration:

NIST Incident Response Framework (SP 800-61)
https://nvlpubs.nist.gov/nistpubs/SpecialPublications/NIST.SP.800-61r2.pdf

Our Playbook: Our modified playbook for The Data Dig: (Can be a WIP, but clear differences should be visible from the Original Playbook)

See the custom playbook here: Custom SolarWinds IOC Investigation Playbook

Description of Changes:

We modified the original NIST playbook to focus specifically on data-driven analysis using Splunk and external threat intelligence tools. Instead of a general incident workflow, our version emphasizes detecting threats by matching IOCs with network logs, validating indicators using VirusTotal, and identifying affected systems quickly. This makes the playbook more practical for log analysis and faster incident triage in a simulated environment.

Stretch Feature: One-Pager (Optional)

One-Pager Draft: Please include a draft of a one-page handout, or "one-pager", you can give your audience prior, during, or after the presentation. One-pagers can be used both to provide extra context and summarize key information.

Note: If you link to Google Docs/Slides/etc, be sure your document is set to *"Anyone with the link can view"*!

SolarWinds Threat Detection Using Network Logs

Milestone Workbook (Optional)

Please use this space to brainstorm, draft, share resources, and otherwise plan out your project!

Submission Checklist

👉*Check off each of the features you have completed.* ***You will only be graded on the features you check off.***

Required Features

- ☑ ~~Select one (or more) open-source Datasets to analyze~~
  - ☑ ~~Data Set Chosen (Name & Link)~~
  - ☑ ~~Data Set Description~~
  - ☑ ~~3 Hypotheses Made~~
- ☑ ~~Select an incident-response playbook to follow~~
  - ☑ ~~Playbook Chosen (Name & Link)~~
  - ☑ ~~Playbook Description~~
  - ☑ ~~2+ Tools Identified~~
- ☑ ~~Answer each of the key aspect questions:~~
  - ☑ ~~Monitoring Sources~~
  - ☑ ~~Identified Assets~~
  - ☑ ~~Impact Analysis and Triage~~
  - ☑ ~~Threat Intelligence~~
  - ☑ ~~Recommended Remediation~~
  - ☑ ~~Case Management System~~
- ☑ ~~Your presentation plan: A roadmap, outline, or diagram~~

Stretch Feature

- ☑ ~~Customize a playbook to fit your dataset / scenario~~
  - ☑ ~~Original/Inspiration Playbook Link~~
  - ☑ ~~Custom Playbook Link~~
  - ☑ ~~Description of Changes~~
- ☑ ~~Submit a draft for a one-pager summarizing your project for the audience~~

💡***Tip: You can see specific grading information, including points breakdown, by going to 🔗 the grading page on the course portal.***

Submit your work!

Step 0: **Decide** which group member will submit! **Only one person should submit the milestone** each unit – So make sure everyone's names/emails are on this document!

Step 1: **Click** the Share button at the top of your screen double check that anyone with the link can edit.



Step 2: **Copy** the link to this document.



Step 3: **Submit** the link on the portal.